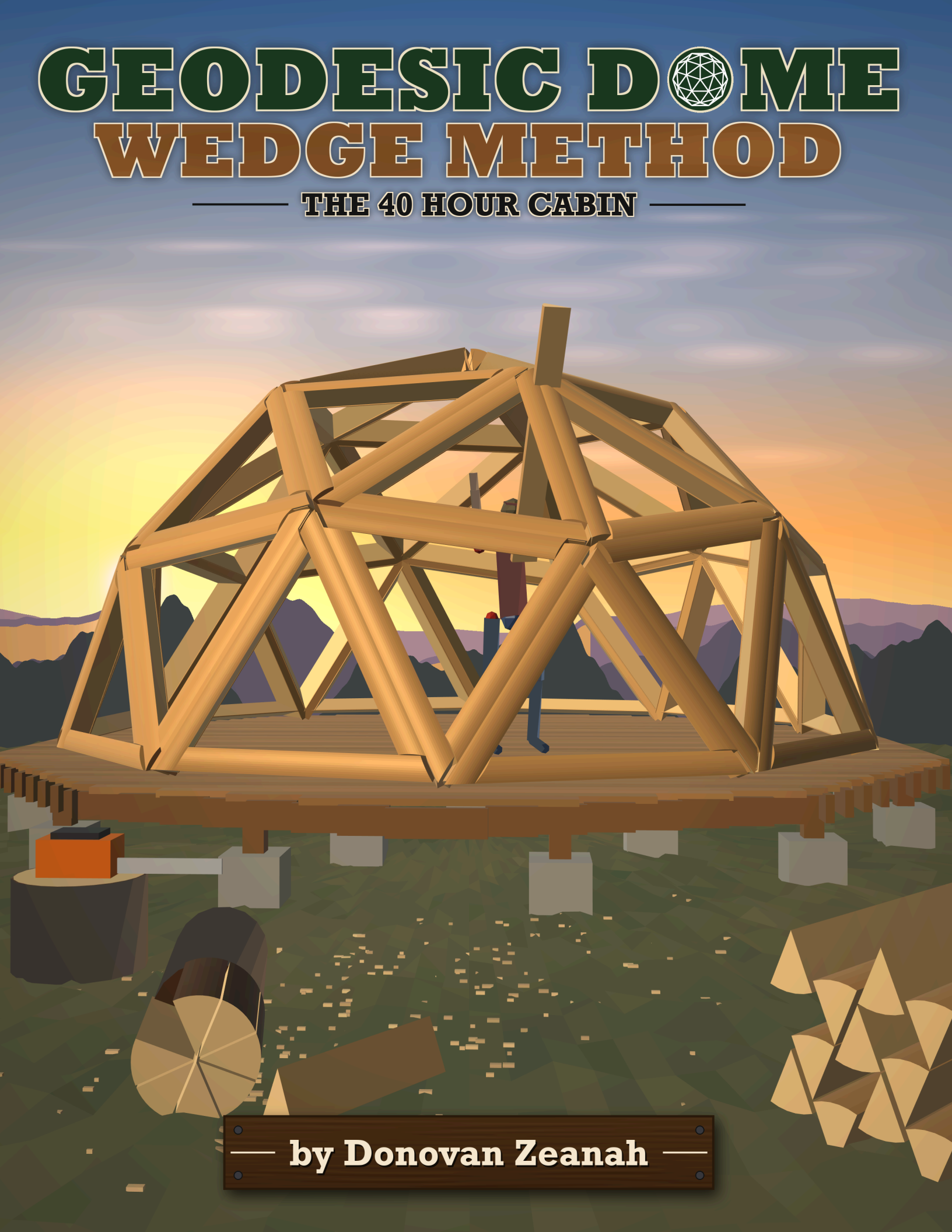
GEODESIC DOME
WEDGE METHOD

THE 40 HOUR CABIN



by Donovan Zeanah

FREE SAMPLE

The 40 Hour Cabin

Complete Edition · Donovan Zeanah

A small timber house, framed from trees you split yourself, with a chainsaw and a jig on one flat board. This book is how, from the first cut to a dry, raised shell -- with every length, angle and price computed from the geometry rather than guessed.

120	40	277 sq ft	37.5 h	$692
split-log members, cut to two lengths	triangular panels, built flat on one jig	of floor under a 19.42-ft dome	of frame work, standing tree to standing frame	for the wedge-cut frame, priced line by line

In this sample

16 of the book's 361 pages, exactly as they are printed, from 4 of its 83 chapters. The page numbers are the book's own.

Forty Hours page 8

1. Your Skull Is Not Symmetrical Either page 12

2. Water Is the Silent Killer page 16

3. Build It So You Can Change Your Mind page 19

13. Stop Squaring. Start Splitting. page 54

The full contents, and where to get the whole book, are at the end.

Forty Hours

The goal is a house built so efficiently that one person can go from standing timber to a standing shell inside a single working week.

Not a kit. Not a shell delivered on a lorry. A tree, a chainsaw, and 40 hours.

That is not an aspiration bolted onto the method afterwards. It is where the number came from, and here is the arithmetic.

The clock

Splitting. One session at the log — 4 hours with a meagre chainsaw — produces 30 wedges. The frame needs 120, so every member in the building is **16 hours** at the log.

Everything else. Taken end to end, one member costs about 15 minutes from tree to standing: split, crosscut, drilled, joined, and up in the shell. That is an average over the fast work and the slow, not a stopwatch on one operation.

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120 members x 15 minutes  =  30 hours
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Splitting is 53 per cent of that, which is why Chapter 22 says the rip is the process worth improving first.

Getting it wrong. Add 25 per cent for the stick that splits off the radius, the panel laid down in the wrong order, and the hour spent looking for the drill.

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30 hours + 25%  =  37.5 hours
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Which is a 40 hour week, with a little in hand.

The honest part

You will not do it in five days.

This is hard physical work. Splitting a log is not typing; you get perhaps four good hours at it before the quality of your swing goes, and the quality of your swing is what keeps a split on the radius. Nobody does forty hours of this linearly, and anybody who tells you they did was counting the hours and not the days.

It comes in **spurts**. A weekend at the log. An evening on the bench. A day when three people turn up and the shell goes from nothing to closed.

That does not weaken the number — it is the reason the number is worth having. Forty hours of work is a thing you can *schedule* against a life: five Saturdays, or two weeks of evenings, or one very determined week off. A building measured in months is a building you never start, because months cannot be fitted into a year that already has things in it.

So: the forty hours is a **budget of effort**, not a calendar. The point of counting it is that it is small enough to count.

And the rest of the book

The shell is a week. The skin and the floor — Chapters 2 and 38, the two parts of this building that decide whether it is still standing in thirty years — are not, and this book spends more pages on them than on the frame for exactly that reason.

The frame is the part you can do in a week. It is also the part you are allowed to get wrong.

The whole building, counted the same way

So that nobody has to find it on the last page: the forty hours is the frame and its shell – 37.5 hours with error allowed for. The whole standard building, counted the same way – frame, panels, shell, utility column and the rest – is **67 hours** of labour, and Chapter 63 shows every one of them. The title is the part one person can do in a week. The rest is the part worth doing slowly.

CHAPTER 1

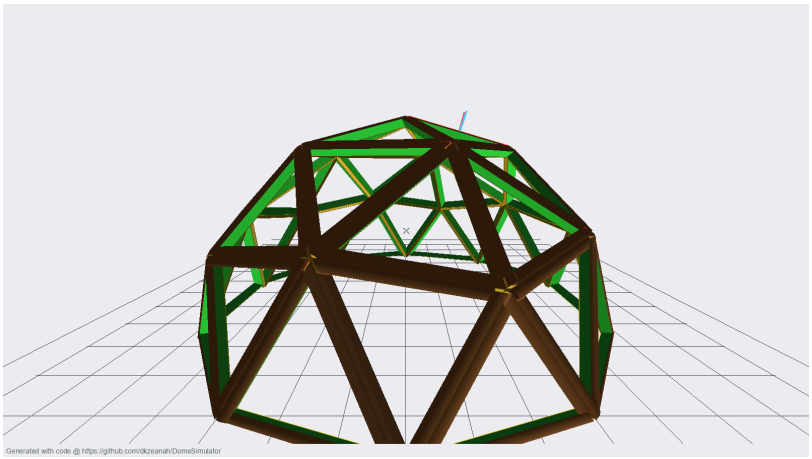
Your Skull Is Not Symmetrical Either

The frame is the part you are allowed to get wrong, and almost every dome book has this backwards

Every dome book I have read opens by telling you to be precise.

Measure twice. Hold your angles. A geodesic dome is a mathematical object and you are about to approximate one out of wood, so the closer you get to the maths the better your building will be.

I believed that for about a year, and it cost me most of a year.



The reference build. Nothing about it is as accurate as it looks.

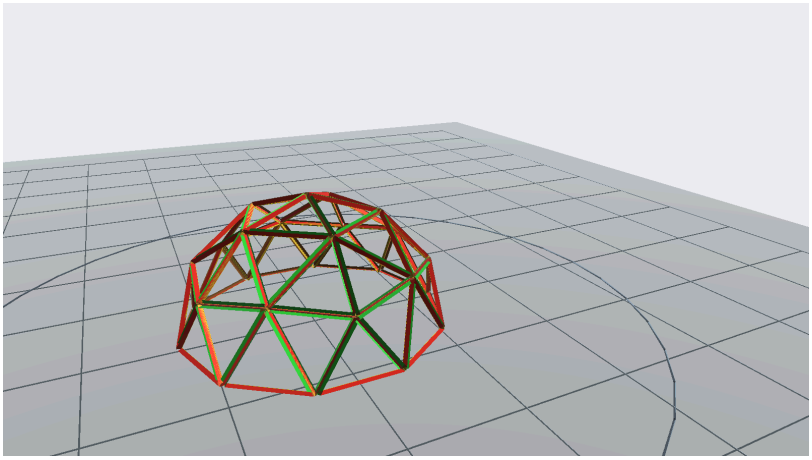
The afternoon I stopped believing it

I had a frame up. Not this one — an earlier one, made of whatever the chainsaw had produced that week, with no two sticks alike. I had built it to prove to myself that the method could not work, so that I could stop thinking about it and go and buy some lumber like a normal person.

It stood. It stood with a stick in it that was out by most of an inch, and a vertex where three members met that I could put my whole thumb into.

I pushed it. I hung off it. I put a ladder against it and climbed on it, which I do not recommend and did anyway.

It did not care.



How much shelter an hour buys, which is the question this book is an answer to.

Why it did not care

A dome carries its loads *axially*. The members are in compression along their own length — pushing end to end, the way a stack of bricks pushes — and a triangulated shell passes a load from one triangle to the next all the way round, so no single member is ever being asked to do very much on its own.

Put a stick in slightly out of position and two things happen. The triangle it is in is very slightly a different triangle. And the load path shifts by that same very slight amount.

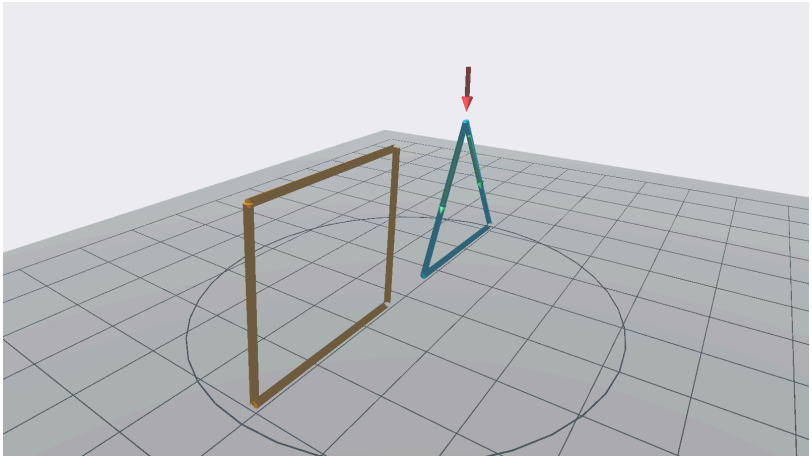
Neither of those is an error that grows. A frame is not a tower of tolerances stacking up toward a failure; it is a shell that absorbs the difference and carries on. The error goes into the seam — and there is already a key in the seam, because there was always going to be one.

Why a triangle, and not a square

Push the top corner of a square frame sideways and it leans. Nothing breaks: the four sides keep their lengths and the corners simply turn. A square is a mechanism, and it needs stiff corners – nailed, braced, glued – to stop being one.

Push the corner of a triangle and nothing moves unless one of its sides gets longer or shorter. Timber is very unwilling to do either along its length. So a triangle needs nothing from its corners at all.

A geodesic shell repeats that triangle across a curved surface, 40 times on this dome. The strength lives in the **network**, not in any one member. No single stick is asked to solve the whole problem; each one is asked to stay its own length, and that is all.



Where the strength actually lives.

That is what lets an odd-shaped member do the job. The structure cares about the member's section, its grain, its unsupported length and the path the load takes through it. It does not care whether the corner of an imaginary rectangle exists – which is the whole reason a split log, with no rectangle anywhere in it, can be a stick in a building.

The lumpy skull

You have an asymmetric skull.

Genuinely: measure it. One side of your head is further from the midline than the other, one eye socket is fractionally lower, your jaw is not centred. This is true of essentially everybody and it is why nobody's face looks right when you mirror it.

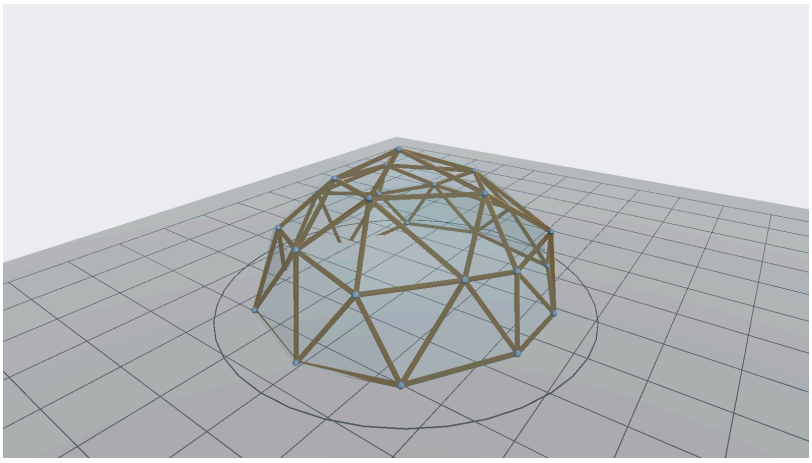
Nobody likes you any less for it, and your skull opposes a baseball bat exactly as well as a perfectly symmetrical one would — because the thing that stops the bat is the *shape*, a closed curved shell in compression, and not the precision.

That is a dome. The geometry does the work. The accuracy is decoration.

There are two reasons the lumpy one stays up, and both are worth knowing before you trust it.

The errors have nowhere to accumulate. A closed, triangulated shell is enormously redundant. Every triangle is rigid on its own, so a joint that is out is absorbed by the triangles around it instead of walking round a ring and adding up the way an error does in a precise, hubbed dome, where every strut has to land on the same point as five others. What you get is a frame that is near the sphere without being on it – a lumpy brain, not a ball.

And the skin covers the rest. One continuous layer bonded over the outside spans the slack between members and gives back the shell action the sloppiness cost. Which is the first hint of the argument the next chapter makes: the frame can be forgiving because the skin is not allowed to be.



The errors have nowhere to accumulate, and the skin covers the rest.

What this book does with that

It spends it.

If the frame does not need precision, then the frame does not need a mill. It does not need a planer, or a moisture-controlled shed, or lumber bought to a tolerance somebody else held for you. It can be made of logs split with a wedge, on the ground, in a field, by one person who is not very good at it yet.

That is the whole method. Everything else in this book — the 45.0-degree sector, the pinwheel joint, the one flat board that makes 40 identical panels — is a consequence of deciding that the frame is the part you are allowed to get wrong.

And what it does not spend

Here is the part that took me another year to learn, and it is the reason this book exists rather than being a pamphlet about splitting logs.

The permission is not general. It applies to the frame and it applies to nothing else.

The skin has to be right, because water does not negotiate and does not announce itself. The floor has to be right, because everything you own sits on it and because the ground sends its water up.

So the advice most dome books give is not merely unhelpful. It is aimed at the wrong layer. It asks for care where care is cheap to skip, and it treats as finishing work the two layers that decide whether the building is still there in thirty years.

The next two chapters are about those two layers. Then we can go and cut something.

CHAPTER 2

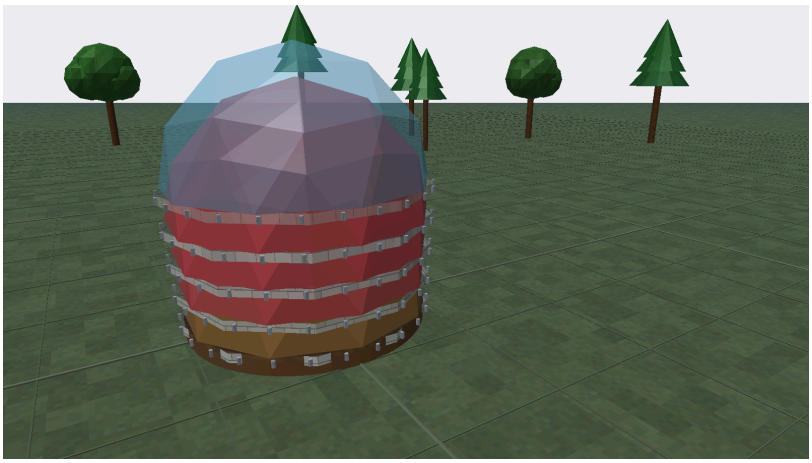
Water Is the Silent Killer

The skin is the part that has to be right, and it is the part every dome book treats as a finish

A frame that is out by an inch stands for fifty years.

A skin that leaks for one winter takes the frame with it.

That asymmetry is the most useful thing in this book, and almost every dome book has it the other way round.



One watertight layer over everything else.

Why water gets to decide

Structural failure is loud. Something cracks, something sags, something you can see from across the room. You get warning, and the warning arrives in a form that makes you do something about it.

Rot is not loud. Rot is a slow chemical argument between a fungus and your building, conducted at about twenty per cent moisture content, entirely out of sight, and it is usually two or three years old by the time anything shows on a surface you look at.

By then you are not repairing a leak. You are replacing members, and the members are the thing this whole method spent its cleverness on making cheap — so you are not even losing much money. You are losing the fortnight.

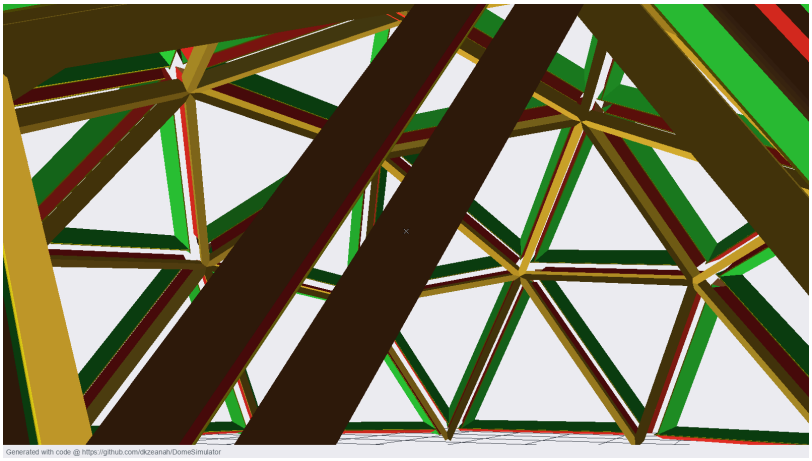
That is the trade nobody states plainly: the frame is cheap to build and expensive to replace, because replacing it means taking the building apart.

Where a dome actually leaks

Not through the panels. A panel is a large flat sheet with no holes in it and it is the easiest part of any building to keep water out of.

Domes leak at seams, at penetrations, and at the apex, in that order — and a 2V dome has a great many seams.

This one has 55 interior seams, which is 309 feet of joint. Every foot of that is a place where two panels meet and where somebody has to have done something deliberate.



Every seam in the dome, and the key in each one.

Three hundred feet of joint is not a detail. It is the single largest risk surface the building has, and it is spread out so evenly that no one part of it looks important enough to worry about.

The move that removes the category

You do not solve three hundred feet of joint by making each joint better.

You solve it by putting one continuous layer over all of it, so that the joints stop being the outside of the building.

That is what the shower cap is. One watertight membrane over the whole shell, strapped down, with nothing structural fastened through it and no seam in it that is not a welded one. The panels underneath do not have to be watertight. The seams underneath do not have to be watertight. They have to be dry, and they are dry because something above them is doing that job and only that job.

The chapter on the cap (28) has the physics and the arithmetic. What matters here is the ordering: **one layer whose only job is water, and it goes on last and comes off first.**

And the floor

The floor is the second thing, and for a different reason.

It is the surface you are in contact with all day. It carries everything you own. It is the only part of the building that is in permanent contact with the ground, which means it is where ground moisture arrives, and unlike a wall it cannot dry to the outside because there is no outside — there is dirt.

A floor that is cold, or damp, or that flexes, is a floor you notice every hour you are awake. A frame that is out of true is a frame you never think about again after the first week.

Get the floor right. Chapter 38 says what right means, and what the five ways of doing it cost on this footprint.

The hierarchy, stated once

1. **The skin.** Water. One continuous layer, no compromises, no penetrations you did not design.
2. **The floor.** Comfort, dryness, and everything you own.

3. **The frame.** Standing up, which it will do whether or not you were careful.

If you read nothing else in this book, invert your instincts in that order. It is worth more than every angle in the appendix.

The catch, and the next chapter

There is an obvious objection, and it is a good one.

The skin and the floor are the expensive parts. They are the parts you cannot cut out of a tree with a chainsaw. Telling somebody that the two layers that matter are also the two layers that cost money is not obviously helpful.

The answer is that they are also the two layers you can *defer* — and the next chapter is about the four things you do while building the frame that keep that option open for the life of the building.

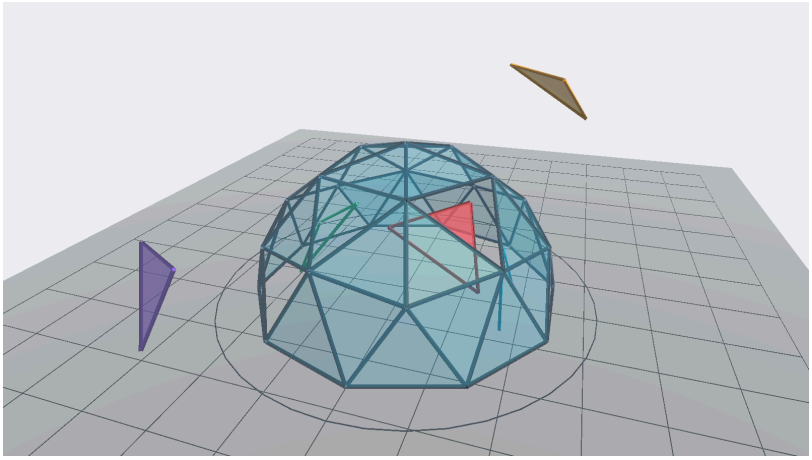
CHAPTER 3

Build It So You Can Change Your Mind

The skin and the floor are the two things that matter and the two things you can defer -- if the frame is built for it

The two layers that matter most are the two layers you can put off.

That sounds like a consolation and it is actually the design. A frame is a commitment: 120 members, cut to two lengths, and once the first panel is closed the shape of the building is decided. A skin is not. A floor is not. Those are things the building *wears*, and a building can change what it wears.



New insulation? Swap it. Solar panels? Add them.

The condition

It only works if you build for it, and the decisions are all made in the first week, before there is a building to look at.

None of them costs anything meaningful at the time. All of them are effectively impossible to add later.

The four things

Threaded inserts in the wedge faces, not screws into end grain.

A panel that will be taken off and put back needs somewhere to bolt to, and end grain is not it — the second time you drive a screw into the same hole it is holding about half of what it held the first time, and the fourth time it is holding nothing. An insert set into the flat sawn face of a wedge is a machine thread in a piece of metal. It does not care how many times you use it.

This is the single highest-leverage decision in the build, and it happens while the panel is still flat on the jig.

A cap sized a layer larger than it needs.

The outer skin goes over whatever is under it, so its size depends on how much is under it. A cap cut to fit a bare shell will not go over a shell wearing two quilted layers, and you find that out on the day you want to add the second layer, in November.

Chapter 30 has the arithmetic — each hat is a size up, and the sequence is not one number. Buy one size up from what you need on day one.

A floor that clamps rather than is built in.

A floor framed into the base ring is part of the structure. A floor that is a hub, radial spokes and decking, clamped to something in the middle, is a component — it can come out, go back, be replaced with a better one, or be left behind.

It also turns out to be the thing that makes a floating dome possible at all, which was not the reason for doing it.

Services in the seam channel, not in the wall.

The seam leaves a gap that has to be filled with something. Fill it with a hose and the building's plumbing is in its structure, reachable from outside by taking off one panel, and no pipe has ever been buried in anything.

An A seam's channel takes a line up to 2.878 inches across. That is not a tight fit for the things a small building actually needs.

What deferring costs

Honestly: some money, later, in exchange for less money now.

The standard article in this book's cost model lists at $12,036, built to $10,030. The shell inside that is $3,265. The bare version of the same cap — no quilted layers at all — is $3,265, and each layer you add afterwards is a few tens of dollars of material and an evening.

So the deferral is real and it is cheap. What you are buying with the four decisions above is the right to spend that money in the order your life actually allows, rather than in the order a building would prefer.

The failure this prevents

I have watched somebody build a beautiful frame, run out of money, put a tarpaulin over it for a winter, and lose the frame.

Not because the tarpaulin was a bad idea. A tarpaulin is a perfectly good temporary skin. They lost it because there was nowhere for a real skin to *attach* when the money came back — no inserts, no strap points, nothing but a frame that had been designed on the assumption that the skin would be figured out later by somebody with more time.

Later is not a plan. Later is a thing you design for.

Every answer is yes

Built that way, the questions people ask about a dome all get the same answer. New insulation? Swap it. Solar? Add it to a panel. A greenhouse wall? Sure. A workshop extension? Bolt it to the rim. Some technology nobody has invented yet? There should be a panel for that – and if the frame is a set of sockets (Chapter 21), there will be.

Where this leaves us

Three chapters in, the argument of this book is complete:

- the frame is allowed to be messy, which is what lets a split log be a structural member;
- the skin and the floor are not, because water and comfort do not forgive;
- and the two layers that are not allowed to be messy are the two you can postpone, upgrade and swap — if you spend four decisions on it in week one.

The rest of the book is how. It starts where the material does, with a tree.

CHAPTER 13

Stop Squaring. Start Splitting.

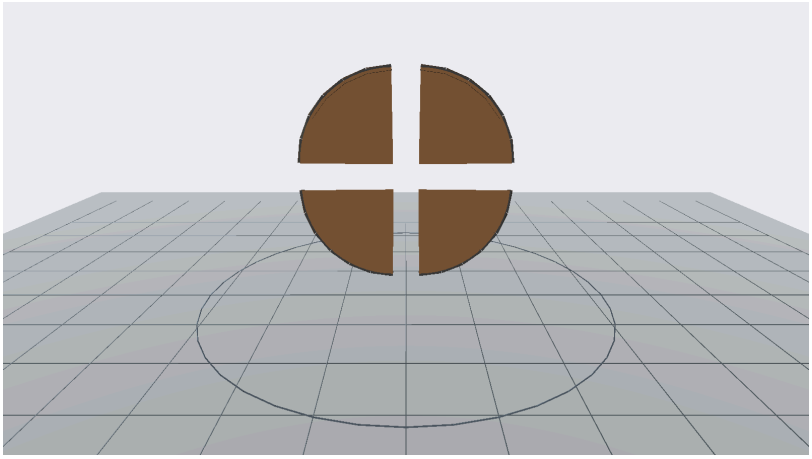
A round log is already rotationally symmetric, and three splits give you eight identical sticks nobody had to measure

I was sitting on a log when this stopped being difficult.

Not thinking about it. Sitting on it, having spent the morning on connectors.

The connector question does have an answer, and it is a flat V bracket: two flaps meeting at the panel's angle, four holes in each flap, and each flap run out to as much as half the member's length so the screws are spread along the stick instead of clustered at its end. Chapter 20 has the detail. It works, it is cheap, and you can make forty of them on a bench brake in an afternoon.

What it does not do is make the sticks alike. It joins dissimilar sticks by absorbing the difference, which means the connector is now doing that work, in every joint, forever.



The same log, opened a different way.

THE 40-HOUR CLOCK

16.0 hours of splitting – the owner's measured rate. The clock goes from 0.0 to **16.0 of 40**, leaving 24.0 hours of the week. Splitting is the biggest single block of the week, and the only one timed at the log rather than worked out.

The wrong question

The question I had been asking was: *how do I join sticks that are all different?*

Everybody's answer to that is the same. Mill them. Square them up, run them through a planer, and now they are all the same because you made them the same.

That answer costs a mill, a planer, a shed to keep the mill dry, the fuel to run it, and — this is the part nobody mentions — about half the tree. A round log does not contain a rectangle. Squaring one is the act of cutting a rectangle out of a circle and calling the rest waste.

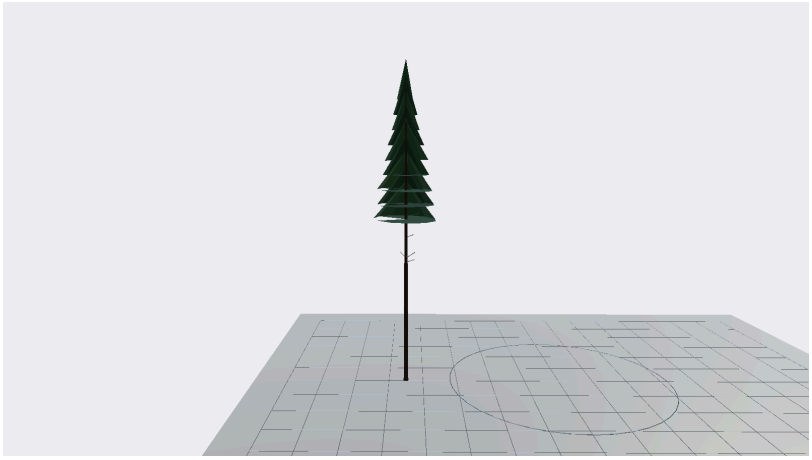
The better question

Sitting on the log, I noticed the log.

A round trunk is rotationally symmetric. Every radius is the same as every other radius. That is not a thing you achieve; it is a thing the tree did while you were not there.

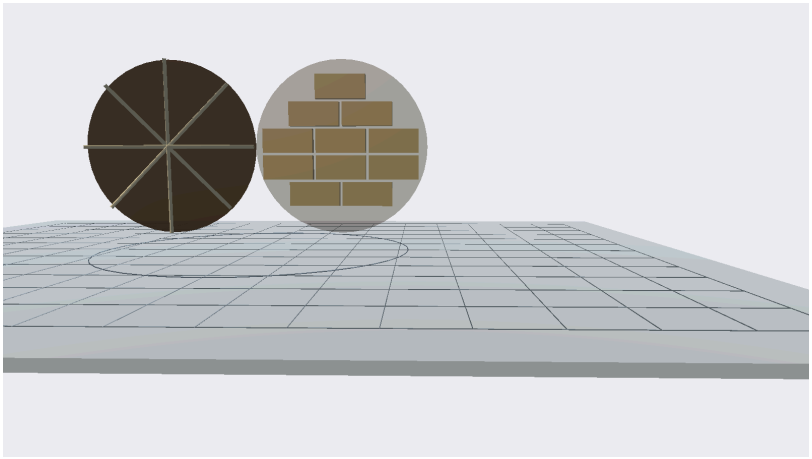
So split it. Split it through the middle and you get two identical halves — identical not because you were careful but because the log was round before you touched it. Split those and you get four. Split those and you get 8.

Three passes. Eight sticks. Every one of them a 45.0-degree sector of the same circle, to whatever tolerance the tree was round to, which is a tolerance nobody had to hold and nobody had to check.



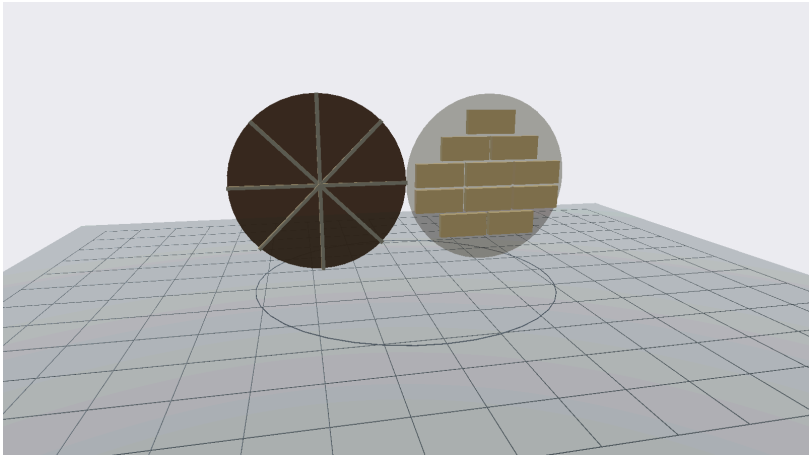
The whole woodpile, standing up.

That was the afternoon. It is not a clever idea and it took me a year to have it, because I was busy asking a question that had no answer.



How much of the stem survives splitting, against sawing.

What the mill is actually for



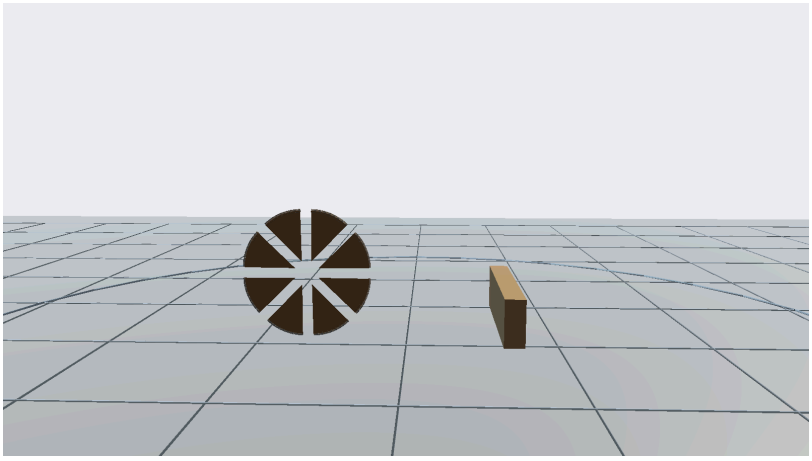
The corners of a round log are not bad wood. They are wood of the wrong shape.

One thing to be careful about, because the recovery argument is easy to overstate.

The corners a mill throws away are not bad wood. They are perfectly good wood of the wrong *shape*. A sawmill exists to turn trees into a standard rectangular product that can be stacked, graded, priced, shipped and sold by people who will never see the tree. It is extremely good at that, and the waste is the price of the standardisation, not of incompetence.

Splitting does not beat sawing at making lumber. It beats sawing at making *this* — a structural member for a frame that does not care about rectangles.

That distinction matters because it tells you when the method stops working. Build something with right angles in it and you want a mill. Build a triangulated shell and you do not.



The tree is the product.

How many rectangles fit in a circle?

Conventional milling squares the log first: four slabs come off the outside to make a rectangular cant. Then the cant is sawn into boards, each board edged, trimmed, dried and graded. Every step removes wood, and every one costs a machine, a setup and a handling. What survives is beautifully standard. What is on the floor is the curved outside of the tree – which is also the densest wood it grew.

How much survives is a packing problem: true two-by-fours, laid out in rows inside each section's small end, where a row can only be as wide as its narrowest edge. For the worked tree of the last chapter:

```
to get true 2 in x 4 in sticks, rectangles have to be packed
into a circle. A mill saws parallel rows, and a row is only as
wide as its narrowest edge, because a stick has to be full
width along its whole length:
  section 1 (13.1 in small end): 0 + 2 + 3 + 3 + 2 + 1 = 11
  section 2 (11.2 in small end): 0 + 2 + 2 + 2 + 1 = 7
  section 3 (9.3 in small end): 0 + 1 + 2 + 1 = 4
  section 4 (7.4 in small end): 0 + 1 + 1 = 2
  section 5 (5.5 in small end): 0 + 1 = 1
  total = 25 true 2x4x12 per tree
  each one holds 8 bf, so 25 x 8 = 200 bf
  out of 442.1 bf standing, that is 45.2% recovery
the brief this lesson works from estimates 32 pieces
  (256 bf, 57.9%),
  which is kinder to the mill than this packing. Both numbers
  are carried through to the comparison.
```

Note what the working does with its own brief. The brief it started from estimated a kinder figure for the mill than honest packing gives, and both are carried forward rather than the convenient one.

The same two trees, both ways

Put the two conversions side by side, for the same two trees:

```
the same two trees, converted both ways:
  radial wedges      782 bf   160 struts
  packed 2x4s        400 bf   100 struts
  ratio = 782 / 400 = x1.95
against the brief's kinder estimate for the mill:
  brief's 2x4s       512 bf   128 struts
  ratio = 782 / 512 = x1.53
  = 53% more wood, which is the figure the brief quotes
either way the direction is the same, and the member count
  moves with it:
  160 wedge struts against 128 sawn ones = 32 more
```

Somewhere between those two ratios more structural wood comes out of the same two trees split than sawn – and one fewer machine was needed to get it.

What this does not claim

One honest paragraph, because the argument does not need overstating.

This is not the claim that any eighth of any tree equals a graded two-by-four. It does not. Species, moisture, knots, checks, decay, grain running off the axis, taper and the connections all still matter, and a building people live in has to be designed and checked around the members actually being used. The figures in this chapter are geometry and volume: what the tree contains and what the frame needs. They are not a structural calculation, and they are not a substitute for one. Chapter 15 is where the structural numbers start.

What comes off the log

At the reference build's 12.0-inch trunk, each sector is 4.592 inches across its bark face and 6.0 inches deep from point to bark. The frame needs 120 of them, which is 664 feet of stick.

Nothing in that sentence required a tape measure on the tree. It required a chainsaw, a steady rip along the radius, and a willingness to let the log decide the section.

The next chapter is what that section actually is, and which way round to put it — which is the first decision in this build you cannot take back.

The round tree is not defective lumber

So, plainly: a wedge from a small tree is not a stronger stick than a stud. It is a cheaper stick, a faster one, one that leaves far more of the tree standing as structure instead of shavings, and one you can make without owning a factory. From a big enough tree it is also simply the stronger stick, and Chapter 15 gives the diameter where that flips.

What makes it enough, either way, is the building. A triangulated shell puts its members mostly in compression along their length, doubles every interior edge, and asks only that the ends can be made to meet. Efficiency is not always a better machine for making a conventional material. Sometimes it is a structure that no longer needs the conventional material at all.

The round tree is not defective lumber. The wedge is not an unfinished two-by-four. It is the member, and the dome is the geometry that lets you use it.

THE WHOLE BOOK

15 Parts, 83 chapters

Chapters in bold are in this sample.

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3 Build It So You Can Change Your Mind
4 A House With a Parts List
5 Why I Built the Ugly One

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15 One Eighth of a Tree, Used as a Stick
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19 Linseed Oil, Where It Belongs

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KEEP READING

Where to get the rest

You have read the part of the book that says what has to be right. The rest is how to make it right: the geometry from one number, the stick, the panel, the skin, the seam, the ground it stands on, what all of it costs, line by line -- and every way to style the finished dome, priced.

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